

MCKELLAR-SIPES, TN, USA

WMO: 723346

Lat: 35.593N

Lon: 88.917W

Elev: 132

StdP: 99.75

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03811

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -7.0	(d) -15.3	(e) 1.0	(f) -6.5	(g) -13.0	(h) 1.2	(i) -5.0	(j) 10.6	(k) 6.7	(l) 9.3	(m) 7.6	(n) 2.7	(o) 20	(p) 0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	34.9	24.9	33.8	24.9	32.6	24.6	26.8	32.2	26.1	31.6	25.5	30.8	3.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.7	29.8	24.6	19.9	29.0	24.0	19.2	28.4	84.2	32.1	81.4	31.6	78.9	30.9	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.8	7.9	7.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-13.5	36.7	2.8	1.7	-15.5	37.9	-17.2	38.9	-18.8	39.9	-20.8	41.2		
			-14.1	27.8	2.8	0.8	-16.1	28.3	-17.7	28.7	-19.2	29.2	-21.2	29.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.1	6.2	10.7	16.0	20.7	24.9	26.4	26.0	22.2	16.1	9.8	5.7	(6)	(7)
	DBStd	9.24	6.59	6.14	5.73	4.89	3.91	2.67	2.27	2.44	3.54	4.75	5.33	5.79	(7)	(8)
	HDD10.0	651	207	137	62	9	0	0	0	0	0	7	68	161	(8)	(9)
	HDD18.3	1913	442	341	244	100	23	1	0	0	9	102	258	393	(9)	(10)
	CDD10.0	2757	23	31	83	189	333	446	507	495	366	194	63	27	(10)	(11)
	CDD18.3	979	1	1	6	31	98	197	249	237	126	31	3	1	(11)	(12)
	CDH23.3	9536	0	2	51	256	852	1946	2547	2394	1172	300	16	1	(12)	(13)
	CDH26.7	3962	0	0	5	44	260	822	1166	1126	469	70	1	0	(13)	(14)
(14) Wind	WSAvg	2.9	3.7	3.7	3.7	3.6	3.0	2.4	2.1	1.9	2.0	2.5	3.0	3.3	(14)	(15)
(15) Precipitation	PrecAvg	1349	95	112	130	134	147	111	106	78	99	82	123	134	(15)	(16)
	PrecMax	1930	222	261	312	288	367	281	297	260	332	188	262	317	(16)	(17)
	PrecMin	881	17	31	44	30	36	14	3	20	11	0	46	22	(17)	(18)
	PrecStd	251	54	60	68	67	75	58	63	51	68	51	55	81	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.2	27.4	29.7	32.4	34.7	36.2	37.5	35.1	30.9	25.5	22.7	(19)	(20)
		MCWB	17.1	17.9	18.3	20.5	23.1	24.3	25.7	24.7	23.0	21.4	18.6	18.3	(20)	(21)
	2%	DB	19.1	21.0	24.8	28.0	30.9	33.6	34.6	35.2	33.0	28.5	22.9	19.6	(21)	(22)
		MCWB	15.9	16.2	16.7	19.1	22.4	24.6	25.7	24.7	23.4	20.7	16.9	16.2	(22)	(23)
	5%	DB	17.2	18.8	22.6	26.4	29.5	32.5	33.5	33.6	31.3	26.9	21.1	17.4	(23)	(24)
		MCWB	14.2	14.2	15.7	18.4	21.6	24.1	25.5	24.9	22.8	19.2	16.0	14.4	(24)	(25)
	10%	DB	14.6	16.6	20.4	24.6	28.1	31.3	32.3	32.2	29.6	24.6	18.9	15.2	(25)	(26)
		MCWB	11.8	12.6	14.5	17.5	21.1	23.7	25.1	24.5	22.0	18.5	14.9	12.6	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	19.0	19.6	22.3	24.8	26.8	27.8	27.7	25.4	24.0	20.5	18.9	(27)	(28)
		MCDB	20.2	21.6	24.6	27.3	29.7	32.2	32.8	33.5	31.4	28.4	22.8	21.4	(28)	(29)
	2%	WB	16.7	17.6	18.2	20.8	23.8	25.9	26.9	26.7	24.6	22.6	18.8	17.0	(29)	(30)
		MCDB	18.8	20.3	23.1	25.7	28.8	31.5	32.3	32.1	30.4	26.4	21.4	18.9	(30)	(31)
	5%	WB	14.8	15.4	16.8	19.8	22.9	25.3	26.3	25.9	23.9	21.2	17.3	15.1	(31)	(32)
		MCDB	16.8	17.9	21.0	24.4	27.8	30.7	31.9	31.2	29.2	24.6	20.0	17.0	(32)	(33)
	10%	WB	12.2	13.2	15.5	18.7	22.0	24.6	25.7	25.3	23.1	19.8	15.4	12.8	(33)	(34)
		MCDB	14.2	15.7	19.7	22.9	26.7	29.5	30.9	30.4	27.5	23.2	18.2	14.7	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.6	11.8	12.2	11.3	11.0	10.7	11.6	12.7	13.0	11.9	9.9	(35)	(36)
		MCDBR	11.3	13.0	13.9	13.6	12.4	12.4	12.0	13.4	14.4	14.8	14.0	11.5	(36)	(37)
	5% WB	MCWBR	8.7	9.1	8.0	6.7	5.6	5.0	4.8	5.2	6.0	7.4	9.0	8.5	(37)	(38)
		MCWBR	9.7	10.8	11.8	10.5	10.6	10.7	10.9	11.2	11.6	11.4	11.0	9.9	(38)	(39)
(39) Clear-Sky Solar Irradiance	taub		0.315	0.325	0.364	0.390	0.430	0.446	0.486	0.492	0.420	0.363	0.329	0.314	(39)	(40)
		taud	2.494	2.455	2.376	2.304	2.250	2.213	2.136	2.106	2.303	2.439	2.481	2.505	(40)	(41)
	Ebn at Noon		887	917	904	894	858	841	805	791	836	862	861	865	(41)	(42)
		Edh at Noon	83	96	114	129	138	143	154	155	121	97	83	78	(42)	(43)
(43) All-Sky Solar Radiation	RadAvg		2.37	2.88	4.00	5.16	5.82	6.43	6.19	5.67	4.83	3.76	2.66	2.00	(43)	(44)
	RadStd		0.20	0.36	0.34	0.34	0.41	0.43	0.34	0.31	0.34	0.46	0.27	0.20	(44)	(45)

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page